

Does AI Boost Grades While Undermining Real Understanding?

The Paradox of Artificial Performance

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Deliverable 4 — Written Report

Visualization format: Interactive Dashboard (Python / Plotly Dash)

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Disclaimer: Generative AI Usage

Generative Artificial Intelligence tools were utilized during the development of this project in compliance with the University of Trento Guidelines. It was used in particular for documents formatting, code refactor and help in debugging. All the generated material has been revised by the authors and final responsibility on the developed material rests with the authors.

1. PURPOSE AND SCOPE OF THIS REPORT

This report answers three questions. First, what we actually built (Section 2). Second, what changed compared to the proposal and why (Sections 3 and 4). Third, what was difficult and how we dealt with it (Section 5). Section 6 closes with a short reflection. We did not treat the proposal as something we had to follow to the letter. We used it more as a record of what we were thinking at the start.

2. THE FINAL SOLUTION

2.1 What we built

The final deliverable is an interactive web dashboard written in Python with Plotly Dash. It is organised in five tabs that take the reader from the broad picture of ChatGPT adoption down to how students themselves feel about AI. At the top there is a banner with four KPI cards that sum up the main message before the reader looks at any chart: 69% global ChatGPT adoption (Dataset 1, $n=23,218$); a paradox gap of $+0.54$ between perceived grade gain and perceived critical-thinking gain at extensive use, against $+0.19$ at rare use; an average grade delta of $+9.83$ points for AI users (Dataset 3, illustrative only); and a perceived utility of 7.44 out of 10 (Survey_AI, $n=91$). Each chart also has a short insight box next to it that explains the takeaway in plain words, so that someone without a statistics background can still follow the story.

Tab	Central question	Primary visualization	Dataset
1 · Overview	How widespread and intensive is ChatGPT use?	Donut, intensity bar, field bar	D1
2 · The Paradox	Does heavier use widen the grades vs critical-thinking gap?	Dual-line + annotated gap; gap-by-field lines	D1
3 · Dependency Lens	Which tasks define intensive use, and how do key metrics evolve?	Task $\times$ intensity heatmap; 3-metric bars	D1
4 · Grade Delta	Does AI adoption produce measurable before/after gains?	Before $\rightarrow$ after dumbbell; delta-by-purpose bars	D3
5 · Perceptions	Are students aware of the trade-off?	Emotion radar, satisfaction bars, feelings, advantages	D2 + D1

Table 1. The five tabs of the dashboard, each answering one question with the most suitable dataset.

2.2 Technical implementation

We split the code into four files, each with a clear job. `config.py` holds the colours, the colour maps and the shared `base_layout()` helper. `data.py` loads the three datasets, builds the derived columns (label mappings, `grade_delta`, the ChatGPT-user sub-sample) and computes the KPI values. `figures.py` has one function per chart, each returning a Plotly figure. `app.py` puts the layout together and starts the server. To make the project easy to run, we added a `requirements.txt`, a `Dockerfile` and a `docker-compose` file, so the dashboard starts with a single command (`docker compose up --build`) on `localhost:8050`, without installing Python locally. For speed, the full 23,218-row dataset is loaded only once when the program starts, and all the figures are computed at startup instead of on every callback, which keeps the interaction fast.

2.3 Design language

The dashboard uses a dark, minimal look with the same colours across all tabs: teal for positive grade signals and AI users, red for the critical-thinking lag, amber for the gap and the concern metrics, and grey for non-users and neutral references. Fields of study get their own four-colour sub-palette so they can be compared in Tab 2. The Likert axes are always shown on the 1–5 scale, so the charts can be read against each other. We also built the tabs so that Tab 1 works on its own for a quick 30-second look, while the later tabs add detail without needing the previous ones.

3. DEVELOPMENT TIMELINE

The project went through five phases. The most important decision, swapping the synthetic dataset for a real survey, was actually taken before the proposal reached its final (v3) version. That is the main reason the proposal and the final dashboard already look so similar.

Phase	What happened	Outcome
Concept & first proposal	We framed the paradox question; at this point the main dataset was a fully synthetic one.	Idea and narrative arc defined
Proposal revision (→ v3)	We replaced the synthetic dataset with the real Aristovnik et al. (2024) global survey and fixed the 5-tab structure, the KPIs, the palette and the wireframe.	Proposal presented (late March)
Technical report (D2)	We ran the EDA on all three datasets, confirmed the paradox gap on real data and documented the missing data, the scales and the biases.	Data quality checked
Implementation (D3)	We built the modular Dash app and adjusted a few charts while turning the EDA into the actual design.	Working dashboard
Written report (D4)	We wrote up the process, the differences and the difficulties.	This document

Table 2. The phases of the project.

4. DIFFERENCES BETWEEN THE PROPOSAL AND THE FINAL SOLUTION

4.1 From synthetic to real data

In an earlier version of the proposal our main dataset was synthetic. We replaced it with the Aristovnik et al. (2024) Global ChatGPT Student Survey, which has 23,218 real answers from 109 countries and was peer-reviewed (PLOS ONE, 2025). This changed the whole project. Instead of only illustrating the paradox, we could now show it on real data, where the gap between perceived grades and perceived critical thinking grows steadily from +0.19 at rare use to +0.54 at extensive use.

Because of this we kept the semi-synthetic Dataset 3 only as an illustrative example, and not as real evidence.

4.2 Changes to single charts

- **Slope chart (Tab 4).** In the proposal we wanted to draw the roughly 100 individual before/after lines, coloured by purpose. When we tried it the chart was unreadable: 100 overlapping lines, many of them flat at $\Delta=0$ for the non-users. So we switched to four group averages (Non-Users, Research, Coding, Homework) shown as before → after dumbbells with the Δ written on them. This makes both the difference and the flat synthetic line easy to see.
- **Grade delta by purpose (Tab 4).** We added $\pm$ SD error bars and the SD value next to each bar, to show that the samples are small (about 13–14 students per purpose). The proposal did not make this uncertainty visible.
- **Satisfaction and attitude (Tab 5).** We did not use the full Q24a–g / Q25a–d set of items. We picked eight of them so the bar chart stays readable.
- **Slope colouring.** The proposal said we would colour by delta_direction (positive/zero). In the end we coloured by group and purpose, which says more once the non-users are already shown as a single flat line.

4.3 Data problems we found during the EDA

- **Gender (Dataset 2).** In the proposal we wrote that the sample was mostly female (64.8%). While doing the EDA we found it was the opposite, 64.8% male and 35.2% female, so we had read the gender variable the wrong way round. We corrected this, and the comment about the STEM skew that came with it.
- **Column names.** One column in the Survey_AI file (Q10.Advantage_evaluation) had a trailing space in its name, which we had to clean when loading the data, otherwise it raised a key error.
- **Different scales.** The advantage/disadvantage questions in Dataset 2 use a 1–3 scale (1 = agree, 3 = disagree), which is not the same as the 1–5 Likert scale in Dataset 1. We adjusted the charts, the reference lines and the titles so that “below the midpoint = agreement” is clear and is not read backwards.

4.4 What stayed the same, and why

The five tabs, the KPI banner, the dark palette, the one-question-per-tab rule and the insight boxes stayed basically the same. This was not because we did not bother to change them. The EDA confirmed that the structure worked: the paradox gap was clean and went in one direction, so it did not need any smoothing, and the dual-line plus gap chart we had planned was enough. The three datasets also told the same story, which is what we had hoped for in the proposal.

5. DIFFICULTIES ENCOUNTERED AND HOW WE ADDRESSED THEM

Most of the real difficulty was not technical but about being honest with the data: deciding what we could actually say, and making sure the reader would not read too much into it.

Difficulty	How we addressed it
The grades and critical-thinking measures (Q27b, Q29e) are self-reported, not objective outcomes.	We always wrote “perceived”, never talked about objective grades, and checked against Dataset 3 where we could.
Dataset 1 is cross-sectional, so we cannot prove cause and effect.	We used correlational wording, kept before/after only for Dataset 3, and said this clearly in the Tab 2 insight box.
Synthetic artifact: the non-user Δ is exactly 0.00 for all 60 students (D3).	We showed the artifact instead of hiding it, with a note on the chart and an “illustrative” label; the dumbbell makes the flat line visible.

Difficulty	How we addressed it
The three datasets cannot be merged at the student level.	Each tab uses one dataset and says which one; we only compare across datasets at the aggregate level.
The scales are different and partly inverted (D1 1–5 vs D2 1–3).	We handled this chart by chart, with neutral reference lines and the scale written in the title.
A lot of values are missing (~42–45% on the user-only questions).	We explained that this is by design, not lost data, and we avoided per-country breakdowns where the groups were too small.
The strongest signal sits in a small group (Extensively, n=694, 4.3%).	We flagged this group in the charts so it is not over-read, even though it carries the clearest paradox.
A counter-intuitive result: the concern about AI drops as use goes up (Q22j).	We gave the two possible explanations (real skill vs cognitive dissonance) and did not claim a cause.
The dataset is large (23,218 rows, 174 columns), so loading is slow.	We load it once at startup and pre-compute all the figures.
Risk of the five tabs feeling overloaded.	One question per tab, the same layout every time, and a single palette.
Making the project run on any machine.	Docker plus a requirements.txt, so it runs with one command.

Table 3. The main difficulties and what we did about them.

6. REFLECTION AND CONCLUSION

The main thing we learned is that the data, more than the design, decided what we could say. Once we moved from synthetic to real data we had to be more careful: the dashboard never says that AI causes the gap, only that heavier use goes together with a wider perceived gap, and the one number that looks like hard before/after evidence is clearly marked as synthetic. The fact that the proposal and the final dashboard are so close tells us that the early plan was reasonable. The real changes happened in the data and in how carefully we presented its limits, with notes, caveats and labels.

The message we want the reader to leave with is the same as in the proposal, but it now rests on real data: AI does not have a single good-or-bad effect on learning. It depends a lot on how, and how much, the tool is used.

7. LINK TO THE GITHUB REPO WITH THE CODE

https://github.com/Teto03/Data_Viz_lab-project